CSE 3031 (Spring 2026) - Homework 1-4: Extracted Questions with Answers

Note: This PDF is generated from the same solved content as HW1_HW4_Questions_Answers.tex.

HOMEWORK 1
Q1: CPU Time factors most influenced by technology/compiler/architect?
A1: Technology -> cycle time; Compiler -> IC; Architect -> CPI (primarily).
Q2: Effective CPI from instruction mix?
A2: CPI = 0.35*1.4 + 0.46*1.0 + 0.16*1.8 + 0.03*1.2 = 1.274.
Q3a: M2 clock to match M1 for P1?
A3a: CPI_M1(P1)=1.6, CPI_M2(P1)=2.0 => f_M2=1*(2.0/1.6)=1.25 GHz.
Q3b: Same clock, equal runs of P1/P2: faster machine?
A3b: Avg CPI M1=1.85, M2=2.0 => M1 faster by about 8.1%.
Q3c: Workload mix for equal performance?
A3c: 20% P1 and 80% P2.
Q4a: Max speedup with 15% sequential?
A4a: 1/0.15 = 6.6667.
Q4b: Processors within 20% of max?
A4b: N >= 23.
Q4c: Processors within 2% of max?
A4c: N >= 278.
Q5a: MTTF for 100 FIT?
A5a: 10^9/100 = 10,000,000 hours.
Q5b: Availability with 24h repair?
A5b: 10,000,000/(10,000,024)=99.99976%.
Q5c: MTTF for 1000 such processors (fail if one fails)?
A5c: FIT_total=100,000 => MTTF=10,000 hours.

HOMEWORK 2
Q1: Effective CPI using Fig.1 (gap/gcc average)?
A1: CPI = f_ALU*1.0 + f_LS*1.4 + f_J*1.2 + f_CB*1.8, with frequencies from Fig.1 average.
Q2a: 3 two-address, 30 one-address, 45 zero-address possible?
A2a: Yes. 3/2^2 + 30/2^7 + 45/2^12 = 0.995361 <= 1.
Q2b: 3 two-address, 31 one-address, 35 zero-address possible?
A2b: No. 3/2^2 + 31/2^7 + 35/2^12 = 1.000733 > 1.
Q2c: Max one-address with 3 two-address + 24 zero-address already?
A2c: 31.
Q3: ISA-class code-size question (uses Figure 2).
A3: Solved in LaTeX with standard stack/accumulator/reg-mem/load-store assumptions and formulas.

HOMEWORK 3
Q1a: Total tag bits?
A1a: Physical addr=14, tag per line=9, lines=8 => total tag bits=72.
Q1b: Min page size to avoid synonym (virtually indexed)?
A1b: 32 bytes.
Q1c: Page table size if page size=64B?
A1c: 2 KB.
Q1d: Why split L1 and unified L2?
A1d: Split L1 lowers latency and avoids I/D conflicts; unified L2 improves capacity sharing.
Q2: Data cache hits/misses for loop?
A2: Misses=1151, hits=897.
Q3a: Condition for smaller cache advantage (100ns penalty)?
A3a: m1 - m2 < 0.003.
Q3b: For 10ns and 1000ns?
A3b: m1-m2 < 0.03 (10ns); m1-m2 < 0.0003 (1000ns).
Q4: TLB/page-table outcomes for trace?
A4: [1:M/F, 5:H/X, 9:M/F, 14:M/F, 10:H/X, 6:M/H, 15:H/X, 12:M/H, 7:M/H, 2:M/F].
Q5: LRU/FIFO page faults and final frames?

A5: LRU faults=7 final {2,3,10,11}; FIFO faults=8 final {3,11,2,10}.

HOMEWORK 4
Q1a: Ideal pipeline speedup?
A1a: 5x.
Q1b: Decrease due to two data hazards?
A1b: stall/instr=0.25 => speedup=4x, decrease=1x (20%).
Q2a-d: Clock cycle times and pipelined speedup?
A2: single-cycle=7ns, multi-cycle=2ns, pipelined=2ns, speedup=3.5x.
Q3: Branch penalty table (deep pipeline)?
A3: Stall-until-resolved: (2,3,3); Predict taken: (2,2,3); Predict untaken: (2,3,0).
Q4: Dependency analysis?
A4: RAW:1->2 (needs stall), 1->3 (forward), 2->5 (no stall); WAR:2->3,2->4,3->4; WAW:1->4.
Q5: Branch prediction sequences?
A5a 2-bit: B1 predictions N N N N N N N N; B2 predictions N N T T T T T T.
A5b (1,1) correlated: B1 predictions N N N T T N N T; B2 predictions N N T T T N T T.

See full detailed derivations and assumptions in HW1_HW4_Questions_Answers.tex.